

Notes: Caramp & Silva (JF 2026)

Monetary Policy and Wealth Effects: The Role of Risk and Heterogeneity

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1 Big Picture: Question, Contribution, and Map

1.1 What is the paper asking?

The paper studies *how monetary policy affects real activity through asset prices* when:

- there is *aggregate disaster risk* (a rare, permanent drop in productivity), and
- there is *heterogeneity in beliefs and MPCs* (optimists vs pessimists, plus hand-to-mouth workers).

A central target is to jointly explain:

- large conditional movements in risk premia (term premia, credit spreads, equity premia) after monetary shocks, and
- nontrivial real effects (output and inflation responses),

without relying on a quantitatively extreme intertemporal-substitution channel alone.

1.2 Core conceptual ingredients

The paper’s logic progresses in four steps:

1. **Disaster risk creates a time-varying price of risk** that enters asset prices and (potentially) consumption decisions.
2. **Belief heterogeneity creates redistribution across savers** (optimists vs pessimists) when discount rates move.
3. **Heterogeneous iMPCs translate redistribution into aggregate demand effects.**
4. **Fiscal backing and government debt determine whether discount-rate changes become net household wealth effects.**

1.3 What I should remember

- The model is continuous time, New Keynesian (Rotemberg), with a Poisson disaster.
- There are three household groups: workers w and two saver types o, p with different subjective disaster intensities $\lambda_o \leq \lambda_p$.
- A single object—the *market-implied disaster probability* λ_t —aggregates saver heterogeneity into asset pricing and the NK block.
- The key “macro wedge” created by belief heterogeneity is $\hat{\lambda}_t \equiv \log(\lambda_t/\lambda)$, which moves the *price of disaster risk* and therefore risk premia.

2 Environment and Assets: The Full Model

2.1 Time, states, and disasters

Time is continuous, $t \in [0, \infty)$. A Poisson process N_t governs disaster arrivals.

- Under the objective measure, disaster arrival intensity is $\bar{\lambda}$.
- Under subjective beliefs, saver type $j \in \{o, p\}$ uses intensity λ_j with $\lambda_o \leq \lambda_p$.

A disaster is absorbing and causes a permanent drop in productivity from A to A^* .

2.2 Agents and population shares

There are three representative groups (masses μ_w, μ_o, μ_p):

- **Workers** (w): supply labor, borrowing constraint binds (hand-to-mouth in baseline).
- **Optimistic savers** (o): lower perceived disaster intensity λ_o .
- **Pessimistic savers** (p): higher perceived disaster intensity λ_p .

2.3 Production and price setting (New Keynesian block)

A continuum of intermediate producers uses labor only and faces Rotemberg price adjustment costs. The NK Phillips curve will arise in equilibrium (equation (5) below).

2.4 Financial assets in the baseline

Short-term nominal bond (riskless in no-disaster state). Nominal rate is i_t .

Long-term nominal government bond. A bond issued at time t has nominal price $Q_{L,t}$ (no-disaster) and pays coupons $e^{-\psi_L(s-t)}$ at all future times $s \geq t$. Let $Q_{L,t}^*$ be its price in the disaster state. Its (no-disaster) return process can be written as

$$dR_{L,t} = \left(\frac{1}{Q_{L,t}} + \frac{\dot{Q}_{L,t}}{Q_{L,t}} - \psi_L \right) dt + \frac{Q_{L,t}^* - Q_{L,t}}{Q_{L,t}} dN_t.$$

Define the conditional (no-disaster) excess return:

$$r_{L,t} \equiv \left(\frac{1}{Q_{L,t}} + \frac{\dot{Q}_{L,t}}{Q_{L,t}} - \psi_L \right) - i_t.$$

Equity (claim on real profits). Let $Q_{E,t}$ be the equity price (no-disaster) and $Q_{E,t}^*$ in disaster. If real profits are Π_t , then

$$dR_{E,t} = \left(\frac{\Pi_t}{Q_{E,t}} + \frac{\dot{Q}_{E,t}}{Q_{E,t}} \right) dt + \frac{Q_{E,t}^* - Q_{E,t}}{Q_{E,t}} dN_t,$$

and the conditional excess return (over the real short rate $i_t - \pi_t$) is

$$r_{E,t} \equiv \left(\frac{\Pi_t}{Q_{E,t}} + \frac{\dot{Q}_{E,t}}{Q_{E,t}} \right) - (i_t - \pi_t).$$

2.5 Government policy: monetary and fiscal

Monetary policy rule. Nominal rate follows

$$i_t = r_n + \phi_\pi \pi_t + u_t, \tag{6}$$

where r_n is the (stationary-equilibrium) natural rate and u_t is the monetary shock.

Fiscal policy and government budget constraint. Government issues long-term nominal debt and pays transfers $T_{j,t}$ to each group $j \in \{w, o, p\}$. A (log-linearized) fiscal backing process τ_t will adjust to satisfy the no-Ponzi condition; see equation (12).

3 Households: Optimization Problems and Euler Equations

3.1 Savers: preferences, discounting, and budget constraints

A type- j saver chooses consumption and portfolio holdings to maximize expected discounted utility with CRRA risk aversion σ . The paper uses an Uzawa-type discounting device to pin down a stationary wealth distribution:

$$\rho_{j,t} = \rho_j \left(\frac{C_{j,t}}{C_{s,t}} \right), \quad C_{s,t} \equiv \frac{\mu_o}{\mu_o + \mu_p} C_{o,t} + \frac{\mu_p}{\mu_o + \mu_p} C_{p,t}.$$

In log-linear form the paper uses:

$$\rho_{j,t} = \rho_j + \sigma \xi (c_{j,t} - c_{s,t}),$$

with $\xi > 0$ governing the speed of mean reversion in relative consumption/wealth among savers.

Let saver net worth be the sum of positions in short bonds, long bonds, and equity:

$$B_{j,t} = B_{j,t}^S + B_{j,t}^L + B_{j,t}^E.$$

The flow budget constraint (in no-disaster times) is

$$dB_{j,t} = \left[(i_t - \pi_t)B_{j,t} + r_{L,t}B_{j,t}^L + r_{E,t}B_{j,t}^E + T_{j,t} - C_{j,t} \right] dt + (B_{j,t}^* - B_{j,t})dN_t,$$

where post-disaster net worth is

$$B_{j,t}^* = B_{j,t} + B_{j,t}^L \frac{Q_{L,t}^* - Q_{L,t}}{Q_{L,t}} + B_{j,t}^E \frac{Q_{E,t}^* - Q_{E,t}}{Q_{E,t}}.$$

Borrowing constraint in baseline: $B_{j,t} \geq 0$.

3.2 Key saver Euler equations and disaster risk premia

The paper's HJB/FOC logic yields the following conditions.

Short-term bond Euler equation.

$$\frac{\dot{C}_{j,t}}{C_{j,t}} = \sigma^{-1}(i_t - \pi_t - \rho_{j,t}) + \frac{\lambda_j}{\sigma} \left[\left(\frac{C_{j,t}}{C_{j,t}^*} \right)^\sigma - 1 \right]. \quad (1)$$

Risk premia for long bonds and equity (disaster exposure $\times$ price of disaster risk).

$$r_{L,t} = \lambda_j \left(\frac{C_{j,t}}{C_{j,t}^*} \right)^\sigma \frac{Q_{L,t} - Q_{L,t}^*}{Q_{L,t}}, \quad (2)$$

$$r_{E,t} = \lambda_j \left(\frac{C_{j,t}}{C_{j,t}^*} \right)^\sigma \frac{Q_{E,t} - Q_{E,t}^*}{Q_{E,t}}. \quad (3)$$

3.3 Workers: hand-to-mouth consumption and labor supply

Workers consume their current resources; in the log-linearized equilibrium their consumption is proportional to output:

$$c_{w,t} = \frac{WN_w}{PY} (w_t - p_t + n_{w,t}) + T'_w(Y) y_t \equiv \chi_y y_t. \quad (7)$$

The coefficient χ_y controls the cyclical behavior of worker income/inequality (and therefore affects the aggregate EIS through heterogeneity in MPCs).

4 Aggregation: Market-Implied Disaster Probability and the NK Block

4.1 Market-implied disaster probability

A central object is the *market-implied* disaster probability λ_t , which aggregates heterogeneous beliefs into a single pricing kernel.

Proposition 1 (Market-implied disaster probability). *Define λ_t by*

$$\lambda_t \equiv \left[\frac{\mu_o C_{o,t}}{\mu_o C_{o,t} + \mu_p C_{p,t}} \lambda_o^{1/\sigma} + \frac{\mu_p C_{p,t}}{\mu_o C_{o,t} + \mu_p C_{p,t}} \lambda_p^{1/\sigma} \right]^\sigma. \quad (4)$$

Then one can construct an aggregate stochastic discount factor that prices both the long-term bond and equity as if investors shared a common disaster intensity λ_t .

4.2 Nonlinear NK Phillips curve and policy rule

With Rotemberg price adjustment costs, the paper writes the nonlinear NKPC as

$$\dot{\pi}_t = \left(i_t - \pi_t + \lambda_t \frac{\eta_t^*}{\eta_t} \right) \pi_t - \frac{\varepsilon}{\varphi A} \left(\frac{W_t}{P_t} - (1 - \varepsilon^{-1}) A \right) Y_t. \quad (5)$$

Monetary policy follows (6).

4.3 Log-linear aggregate dynamics: Euler equation and NKPC

Let lower-case variables denote log deviations from the stationary equilibrium. Aggregating saver Euler equations yields:

$$\dot{c}_{s,t} = \sigma^{-1} (i_t - \pi_t - r_n) + \frac{\lambda}{\sigma} \left(\frac{C_s}{C_s^*} \right)^\sigma p_{d,t}, \quad (8)$$

where the *price of disaster risk* is

$$p_{d,t} \equiv \sigma (c_{s,t} - c_{s,t}^*) + \hat{\lambda}_t, \quad \hat{\lambda}_t \equiv \log(\lambda_t / \lambda). \quad (9)$$

Proposition 2 (Aggregate dynamics given $(i_t, p_{d,t})$). *Given paths $\{i_t, p_{d,t}\}_{t \geq 0}$, output and inflation satisfy:*

$$\dot{y}_t = \tilde{\sigma}^{-1} (i_t - \pi_t - r_n) + \chi_{pd} p_{d,t}, \quad (10)$$

$$\dot{\pi}_t = \rho \pi_t - \kappa y_t, \quad (11)$$

where

$$\tilde{\sigma}^{-1} \equiv \frac{1 - \mu_w}{1 - \mu_w \chi_y} \sigma^{-1}, \quad \chi_{pd} \equiv \frac{\lambda}{\tilde{\sigma}} \left(\frac{C_s}{C_s^*} \right)^\sigma, \quad \rho \equiv \rho_s + \lambda, \quad \kappa \equiv \varphi^{-1} (\varepsilon - 1) \phi Y.$$

Equivalently, using $p_{d,t} = \tilde{\sigma} y_t + \hat{\lambda}_t$ (with $y_t^* = 0$), one can write:

$$\dot{y}_t = \tilde{\sigma}^{-1} (i_t - \pi_t - r_n) + \delta y_t + \chi_{pd} \hat{\lambda}_t, \quad \delta \equiv \lambda \left(\frac{C_s}{C_s^*} \right)^\sigma.$$

4.4 Fiscal backing: log-linearized government budget

The log-linearized flow budget constraint is

$$\bar{d}_G \dot{d}_{G,t} = i_L \bar{d}_G d_{G,t} + \bar{d}_G (i_t - \pi_t + r_{L,t} - i_L) - (\chi_\tau y_t + \tau_t), \quad (12)$$

where $\bar{d}_G \equiv D_G/Y$ is government debt-to-output in steady state, $\chi_\tau y_t + \tau_t$ is the primary surplus, and τ_t is the endogenous fiscal backing component needed for the no-Ponzi condition.

5 Asset Pricing and the Market-Implied Probability: Key Closed Forms

5.1 Linearized asset pricing with the price of disaster risk

Given (9), the paper expresses bond and equity prices in discounted-integral form.

Long-term bond price at $t = 0$.

$$q_{L,0} = - \int_0^\infty e^{-(\rho+\psi_L)t} (i_t - r_n) dt - \int_0^\infty e^{-(\rho+\psi_L)t} r_L p_{d,t} dt. \quad (13)$$

The first term is the expectations-hypothesis component; the second is a term premium driven by $p_{d,t}$.

Equity price at $t = 0$.

$$q_{E,0} = \frac{Y}{Q_E} \int_0^\infty e^{-\rho t} \hat{\Pi}_t dt - \int_0^\infty e^{-\rho t} (i_t - \pi_t - r_n + r_E p_{d,t}) dt, \quad (14)$$

where

$$\hat{\Pi}_t \equiv y_t - \frac{WN}{PY} (w_t - p_t + n_t)$$

is the (log-linear) profit component. Equity moves via dividends and via discount rates (real rate + risk premia).

5.2 Log-linear link between $\hat{\lambda}_t$ and redistribution among savers

Log-linearizing (4) yields:

$$\frac{1}{\sigma} \lambda^{-1/\sigma} \hat{\lambda}_t = \mu_{c,o} \mu_{c,p} (\lambda_p^{1/\sigma} - \lambda_o^{1/\sigma}) (c_{p,t} - c_{o,t}), \quad (15)$$

where

$$\mu_{c,j} \equiv \frac{\mu_j C_j}{\mu_o C_o + \mu_p C_p}, \quad j \in \{o, p\}.$$

Thus $\hat{\lambda}_t$ rises when the shock shifts consumption (and hence consumption shares) toward pessimists.

6 Approximate Block-Recursivity and the Dynamics of $\hat{\lambda}_t$

6.1 Relative consumption mean reversion

Taking differences in saver Euler equations and using Uzawa discounting implies:

$$\dot{c}_{p,t} - \dot{c}_{o,t} = -\xi(c_{p,t} - c_{o,t}). \quad (16)$$

Proposition 3 (Approximate block-recursivity). *Assume the effect of aggregate saver consumption on risky-asset returns is second order:*

$$r_k \sigma c_{s,t} = \mathcal{O}(\|i_t - r_n\|^2), \quad k \in \{L, E\}.$$

Then $\hat{\lambda}_t$ is approximately independent of (y_t, π_t) and evolves as

$$\hat{\lambda}_t = e^{-\psi_\lambda t} \hat{\lambda}_0, \quad \psi_\lambda = \xi, \quad (17)$$

and, if $i_t - r_n = e^{-\psi_m t}(i_0 - r_n)$, the initial jump satisfies

$$\hat{\lambda}_0 = \varepsilon_\lambda(i_0 - r_n), \quad \varepsilon_\lambda \geq 0, \text{ with strict inequality if } \lambda_p > \lambda_o. \quad (18)$$

6.2 Implication: a parsimonious price-of-risk representation

Combining (9) with (17) yields:

$$p_{d,t} = \tilde{\sigma} y_t + e^{-\psi_\lambda t} \hat{\lambda}_0. \quad (19)$$

7 Wealth Effects, Risk-Premium Neutrality, and Heterogeneous iMPCs

7.1 Household intertemporal budget constraint (IBC)

Let η_t denote the aggregate stochastic discount factor. Household j satisfies:

$$\mathbb{E}_0 \left[\int_0^\infty \frac{\eta_t}{\eta_0} C_{j,t} dt \right] = B_{j,0} + \mathbb{E}_0 \left[\int_0^\infty \frac{\eta_t}{\eta_0} T_{j,t} dt \right]. \quad (20)$$

7.2 Consumption claim pricing and the definition of wealth effect

Define the (date- t) value of a claim to household consumption:

$$Q_{C_j,t} \equiv \mathbb{E}_t \left[\int_t^\infty \frac{\eta_z}{\eta_t} C_{j,z} dz \right].$$

Linearizing yields the pricing condition

$$q_{C_j,0} = \frac{C_j}{Q_{C_j}} \int_0^\infty e^{-\rho t} (c_{j,t} + \chi_{c_j} c_{j,t}^*) dt - \int_0^\infty e^{-\rho t} (i_t - \pi_t - r_n + r_{C_j} p_{d,t}) dt, \quad (21)$$

where $\chi_{c_j} \equiv \frac{\delta}{r_n^*} \frac{C_j^*}{C_j}$ and

$$r_{C_j} \equiv \lambda \left(\frac{C_s}{C_s^*} \right)^\sigma \frac{Q_{C_j} - Q_{C_j}^*}{Q_{C_j}}.$$

Wealth effect (definition). Define household j 's *wealth effect* $\Omega_{j,0}$ as:

$$\Omega_{j,0} \equiv \underbrace{\frac{1}{C_j} (B_j^E q_{E,0} + B_j^L q_{L,0} + Q_{T_j} q_{T_j,0})}_{\text{asset revaluation effect}} + \underbrace{\frac{Q_{C_j}}{C_j} \int_0^\infty e^{-\rho t} (i_t - \pi_t - r_n + r_{C_j} p_{d,t}) dt}_{\text{discount-rate effect on the consumption claim}},$$

where $Q_{T_j,t}$ is the value of the transfer claim,

$$Q_{T_j,t} \equiv \mathbb{E}_t \left[\int_t^\infty \frac{\eta_z}{\eta_t} T_{j,z} dz \right].$$

Then the IBC in log-linear form is:

$$\int_0^\infty e^{-\rho t} (c_{j,t} + \chi_{c_j} c_{j,t}^*) dt = \Omega_{j,0}. \quad (22)$$

7.3 A special cancellation: “consumption equals dividends”

If a household's consumption claim is exactly its asset claim (e.g., consumption equals equity dividends, no transfers), then the asset-revaluation effect and consumption-claim discount-rate effect cancel:

$$\Omega_{j,0} = -\frac{B_j^E}{C_j} \int_0^\infty e^{-\rho t} (i_t - \pi_t - r_n + r_{EP} p_{d,t}) dt + \frac{Q_{C_j}}{C_j} \int_0^\infty e^{-\rho t} (i_t - \pi_t - r_n + r_{C_j} p_{d,t}) dt = 0. \quad (23)$$

This illustrates why large asset-price moves need not translate into large consumption moves.

7.4 Risk-premium neutrality for the aggregate (policy experiment)

Proposition 4 (Risk-premium neutrality via taxes). *Suppose the government levies a consumption tax on savers and rebates revenues back to savers so that the saver Euler equation is insulated from movements in $\hat{\lambda}_t$ (formally: choose the tax so that the $\hat{\lambda}_t$ term in the aggregate Euler equation is exactly offset). Then (y_t, π_t, i_t) are independent of $\hat{\lambda}_t$. Moreover, under specific conditions on government debt maturity/issuance, fiscal backing can also be made independent of $\hat{\lambda}_t$.*

Remark. Figure 1 (discussed later) illustrates this neutrality: output can be unchanged even when bond yields and stock prices respond very differently because risk premia move.

7.5 Redistribution and iMPCs: the key heterogeneous-MPC channel

Lemma 1 (Intertemporal MPCs (iMPCs)). *For saver $j \in \{o, p\}$, define the iMPCs at horizon t with respect to a wealth shock $\Omega_{j,0}$:*

$$\begin{aligned}\mathcal{M}_{j,t} &\equiv \frac{1}{C_j} \frac{\partial C_{j,t}}{\partial \Omega_{j,0}} = \frac{\rho + \xi}{1 + \bar{\chi} \lambda_j^{1/\sigma}} e^{-\xi t}, \\ \mathcal{M}_{j,t}^* &\equiv \frac{1}{C_j^*} \frac{\partial C_{j,t}^*}{\partial \Omega_{j,0}} = \frac{(\rho + \xi) \bar{\chi}^* \lambda_j^{1/\sigma}}{1 + \bar{\chi} \lambda_j^{1/\sigma}} e^{-\xi t},\end{aligned}\tag{24}$$

where $\bar{\chi}$ and $\bar{\chi}^*$ are positive constants. They satisfy the accounting identity:

$$\int_0^\infty e^{-\rho t} \left[\mathcal{M}_{j,t} - \frac{\delta}{r_n^*} \mathcal{M}_{j,t}^* \right] dt = 1. \tag{24'}$$

Proposition 5 (How redistribution generates $\hat{\lambda}_t$). *The market-implied disaster probability responds according to*

$$\hat{\lambda}_t = \bar{\chi}_\lambda (\mathcal{M}_{o,t} - \mathcal{M}_{p,t}) (\Omega_{p,0} - \Omega_{o,0}), \tag{25}$$

for a positive constant $\bar{\chi}_\lambda$. Hence, $\hat{\lambda}_t$ is large when (i) wealth effects differ across saver types and (ii) their iMPCs differ.

8 Closed-Form Output and Inflation Decompositions

Assumption 1 (Local uniqueness condition).

$$\tilde{\sigma}^{-1} \kappa > \rho \delta.$$

Proposition 6 (Aggregate output decomposition in D-HANK). *Assume $i_t - r_n = e^{-\psi_m t} (i_0 - r_n)$ and $\psi_k \neq -\omega$ for $k \in \{m, \lambda\}$. Then output admits the decomposition*

$$y_t = \underbrace{\tilde{\sigma}^{-1} \hat{y}_{m,t}}_{\text{intertemporal substitution effect (ISE)}} + \underbrace{\chi_\lambda \hat{y}_{\lambda,t}}_{\text{time-varying precautionary (TVP)}} + \underbrace{(\rho - \omega) e^{\omega t} \Omega_0}_{\text{GE factor} \times \text{aggregate wealth effect}}, \tag{26}$$

where $\chi_\lambda \equiv \chi_{pd} \varepsilon_\lambda$,

$$\Omega_0 \equiv \sum_{j \in \{w, o, p\}} \mu_j \frac{C_j}{Y} \Omega_{j,0},$$

and for $k \in \{m, \lambda\}$,

$$\hat{y}_{k,t} = \frac{(\rho - \omega) e^{\omega t} - (\rho + \psi_k) e^{-\psi_k t}}{(\bar{\omega} + \psi_k)(\omega + \psi_k)} (i_0 - r_n). \tag{27}$$

Proposition 7 (Inflation decomposition). *Under the same conditions,*

$$\pi_t = \underbrace{\tilde{\sigma}^{-1}\hat{\pi}_{m,t}}_{ISE \text{ component}} + \underbrace{\chi_\lambda \hat{\pi}_{\lambda,t}}_{TVP \text{ component}} + \underbrace{\kappa e^{\omega t} \Omega_0}_{wealth-effect \text{ component}}, \quad (28)$$

where for $k \in \{m, \lambda\}$,

$$\hat{\pi}_{k,t} = \frac{\kappa(e^{\omega t} - e^{-\psi_k t})}{(\omega + \psi_k)(\bar{\omega} + \psi_k)}(i_0 - r_n), \quad \hat{\pi}_{k,0} = 0.$$

8.1 Aggregate wealth effect pinned by fiscal backing

A key closed form is:

$$\Omega_0 = \frac{\rho - \omega}{(\rho - \omega)\chi_\tau + \bar{d}_G \kappa} \left[\int_0^\infty e^{-\rho t} \Delta B_t^L (i_t - r_n + r_L \hat{\lambda}_t) dt - \bar{d}_G \int_0^\infty e^{-\rho t} \hat{\pi}_t dt - \int_0^\infty e^{-\rho t} \tau_t dt \right], \quad (29)$$

where $\hat{\pi}_t \equiv \tilde{\sigma}^{-1}\hat{\pi}_{m,t} + \chi_\lambda \hat{\pi}_{\lambda,t}$, and $\Delta B_t^L \equiv (1 - e^{-\psi_L t})\bar{d}_G$. A central message is: *discount-rate changes affect Ω_0 primarily through government debt and fiscal backing, not through equity.*

9 Extension: Long-Term Risky Household Debt

9.1 Household debt contract and worker consumption

The paper extends the model by allowing workers to borrow via long-term debt. In log-linear form worker consumption becomes

$$c_{w,t} = \chi_y y_t - \left(\frac{\psi_P}{i_P + \psi_P} (i_{P,t} - i_P) - \pi_t \right) \bar{d}_P, \quad (30)$$

where $\bar{d}_P \equiv D_P/Y$ is the (steady-state) household-debt-to-income ratio and $i_{P,t} \equiv Q_{P,t}^{-1} - \psi_P$ is the yield on the private bond.

Proposition 8 (Output with long-term risky household debt). *Assume $i_t - r_n = e^{-\psi_m t}(i_0 - r_n)$ and the same second-order condition as in Proposition 3. Then output is*

$$y_t = \tilde{\sigma}^{-1} \hat{y}_{m,t} + \chi_\lambda \hat{y}_{\lambda,t} + \frac{\mu_w \bar{d}_P \psi_P}{1 - \mu_w \chi_y} \left[\frac{\tilde{\psi}_m \hat{y}_{m,t}}{\rho + \psi_P + \psi_m} + \frac{r_P \varepsilon_\lambda \tilde{\psi}_\lambda \hat{y}_{\lambda,t}}{\rho + \psi_P + \psi_\lambda} \right] + (\rho - \omega) e^{\omega t} \Omega_0, \quad (P8)$$

where $\tilde{\psi}_k \equiv \psi_k + \rho - r_n$ for $k \in \{m, \lambda\}$.

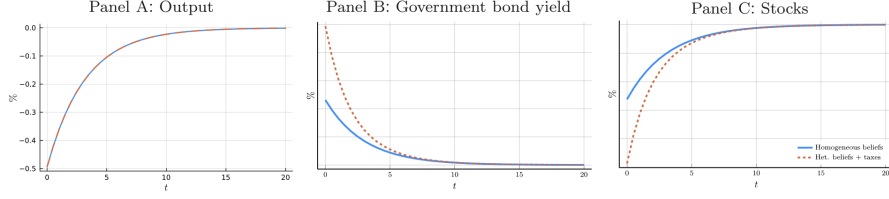


Figure 1. Output and asset prices response. (Color figure can be viewed at wileyonlinelibrary.com)

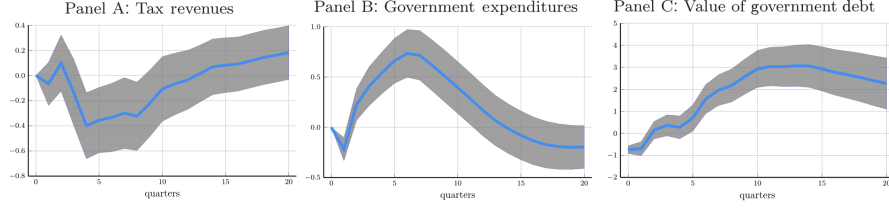


Figure 2. Estimated fiscal response to a monetary policy shock. This figure plots IRFs

10 Empirical/Quantitative Block: What the Figures Show

10.1 Important note about tables

The paper contains no numbered tables. Quantitative evidence is summarized in Figures 1–8.

10.2 Figure 1: Neutral output with different asset-price responses

What is plotted. Three impulse responses comparing “Homogeneous beliefs” vs “Heterogeneous beliefs + taxes”:

- Panel A: output,
- Panel B: government bond yield,
- Panel C: stock market value.

Interpretation. The policy experiment implements Proposition 4: the tax/rebate scheme neutralizes the effect of belief-driven risk-premium movements on aggregate dynamics. Hence output can be essentially identical, even though bond yields and equity prices move differently because term premia and equity premia respond when $\hat{\lambda}_t$ moves.

10.3 Figure 2: Estimated fiscal response to monetary shocks

What is plotted. VAR-based impulse responses of: primary surplus components (revenues and expenditures), PV of interest payments, market value of government debt, and PV of primary surpluses.

Interpretation. Monetary tightening increases the PV of interest payments and lowers the market value of outstanding debt. The empirical fiscal backing is present but modest: the PV of

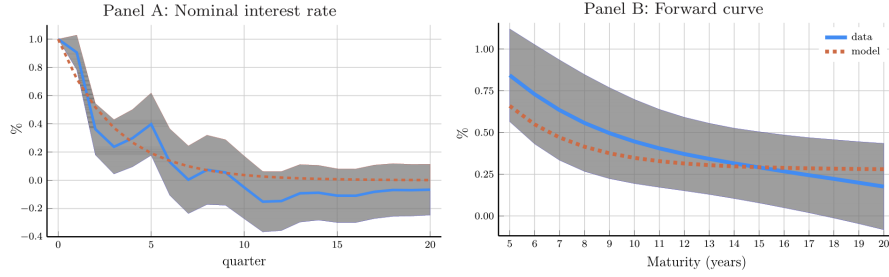


Figure 3. Nominal interest rate and forward curve. Panel A shows the IRF for the federal funds rate in the VAR and the path of nominal interest rates in the model. Panel B shows the response of forward rates to a 100 bps change in the two-year yield, as estimated by Hanson and Stein (2015), and the corresponding forward curve in the model when the monetary shock is scaled such that the two-year yield increases by 100 bps. Gray areas are confidence bands. (Color figure can be viewed at wileyonlinelibrary.com)

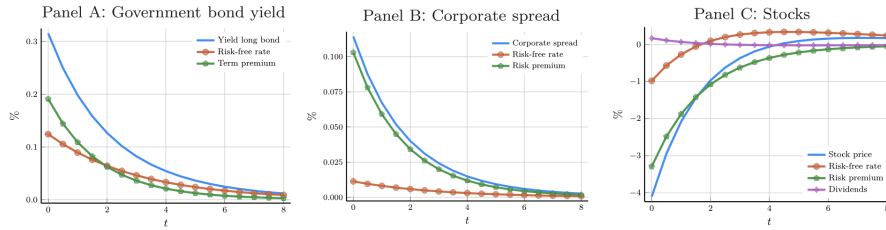


Figure 4. Asset pricing response to monetary shocks. (Color figure can be viewed at wileyonlinelibrary.com)

primary surpluses increases only partially relative to debt revaluation/interest costs. This matters because τ_t and debt revaluation jointly determine Ω_0 in (29).

10.4 Figure 3: Forward curve response and calibration of risk-premium dynamics

What is plotted.

- Panel A: VAR funds rate response vs model path $i_t - r_n = e^{-\psi_m t}(i_0 - r_n)$.
- Panel B: response of forward rates to a change in the two-year yield (empirical estimate vs model).

Interpretation. The model matches not only near-term rate movements but also medium-horizon forward-rate responses by allowing $\hat{\lambda}_t$ to move (hence term premia move). In standard NK/RANK models, far-forward rates tend to respond too little because risk premia are absent or too rigid.

10.5 Figure 4: Asset pricing channels (term premium, risk premium, and dividends)

What is plotted.

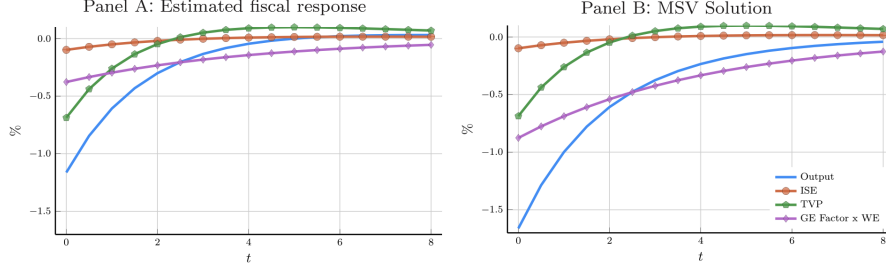


Figure 5. Output in D-HANK. In both graphs, the path of the nominal interest rate is given by $i_t - r_n = e^{-\psi_m t}(i_0 - r_n)$, where $i_0 - r_n$ equals 100 bps. Panel A shows the solution with the estimated fiscal backing, while Panel B shows the MSV solution. (Color figure can be viewed at wileyonlinelibrary.com)

- Panel A: long-term government bond yield decomposed into expected short-rate path vs term premium.
- Panel B: corporate spread decomposed into a default component vs a risk-premium component.
- Panel C: stock market response decomposed into dividend vs discount-rate components, highlighting equity risk premia.

Interpretation. After a tightening, yields rise with a sizable term-premium component, corporate spreads widen partly because risk premia rise, and equity prices fall mostly through discount rates (risk premia), while dividends are comparatively less procyclical in the short run. This joint pattern is a key empirical motivation for putting risk premia into the monetary transmission mechanism.

10.6 Figure 5: Output decomposition in D-HANK

What is plotted. Output response and its decomposition into:

- ISE (intertemporal substitution effect),
- TVP (time-varying precautionary motive),
- GE-adjusted wealth effect (the $(\rho - \omega)e^{\omega t}\Omega_0$ term).

The figure compares an empirically estimated fiscal backing vs an MSV solution that implies much stronger fiscal backing.

Interpretation. With empirically estimated fiscal backing, the model produces a sizable output drop on impact, with TVP contributing the largest share, wealth effects contributing materially, and the pure ISE relatively small. Under the MSV solution, output drops more, mainly because implied fiscal backing makes the aggregate wealth effect larger. This highlights that matching real effects is not only about the NK block but also about fiscal backing and debt revaluation.

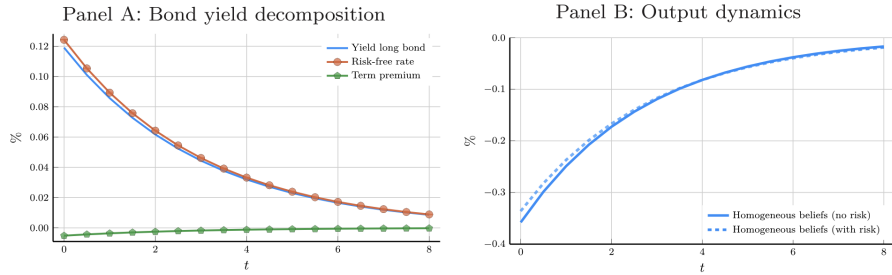


Figure 6. Long-term bond yields and output for economies with and without risk. (Color figure can be viewed at wileyonlinelibrary.com)

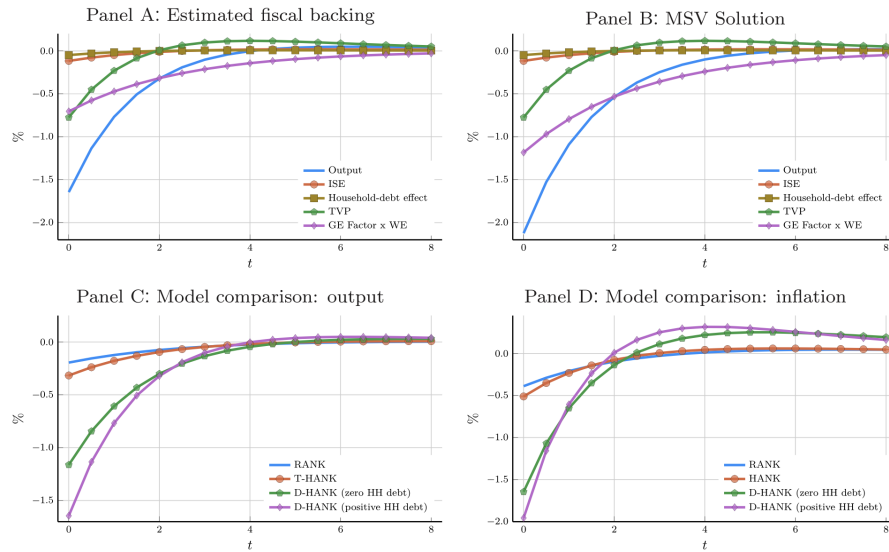


Figure 7. The role of household debt: Output decomposition and model comparison.

10.7 Figure 6: Why heterogeneity matters (risk vs no risk)

What is plotted. Comparison of economies with and without belief-driven risk-premium movements:

- Panel A: yield decomposition with/without term premium response,
- Panel B: output dynamics.

Interpretation. Without belief heterogeneity (so $\hat{\lambda}_t \approx 0$), term premia barely move and output responses are much smaller. Disaster *risk alone* is not enough; *heterogeneity that changes the market-implied probability* is what creates sizable time-varying precautionary forces and realistic term-premium behavior.

10.8 Figure 7: Adding household debt amplifies transmission

What is plotted. Output decomposition and model comparisons when workers have long-term debt:

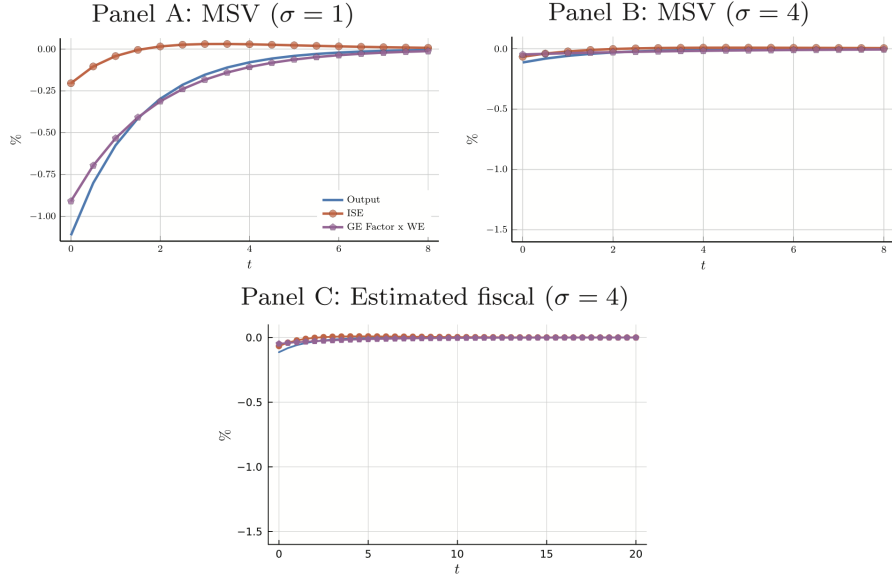


Figure 8. The role of the EIS in RANK's quantitative performance. (Color figure can be viewed at wileyonlinelibrary.com)

- Panels A–B: decomposition under estimated fiscal backing vs MSV.
- Panels C–D: output and inflation comparisons across models (RANK/HANK/D-HANK variants).

Interpretation. Household debt introduces an additional channel (via real debt revaluation and debt-service costs) that amplifies real effects, especially when inflation and private yields move. Importantly, the debt channel interacts with risk and heterogeneity rather than replacing them: the largest effects occur when both (i) risk premia move and (ii) worker debt makes redistribution and real balance-sheet effects stronger.

10.9 Figure 8: The role of EIS in RANK performance

What is plotted. RANK model impulse responses under different EIS calibrations and different fiscal/backing implementations.

Interpretation. To replicate large output drops in RANK, one typically needs either (i) very high EIS (low σ) or (ii) extremely strong fiscal backing/wealth effects. With more conservative EIS values, the intertemporal-substitution channel alone becomes too weak. This is the paper's quantitative argument for why adding risk and heterogeneity is not cosmetic: it helps explain large asset-price movements and nontrivial real effects without forcing extreme EIS assumptions.

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11 Summary

The paper rethinks the *wealth-effects / asset-price channel* of monetary policy by integrating two ingredients that are central for asset prices but often simplified away in standard monetary models:

1. **Risk:** a rare-disaster component that generates sizable and time-varying risk premia (term premia, credit spreads, equity premia).
2. **Heterogeneity:** households differ in beliefs (optimists vs pessimists) and in marginal propensities to consume (high-MPC workers vs low-MPC savers).

The goal is to explain why monetary policy shocks move *long yields, spreads, and equity prices* in empirically realistic ways and how these financial responses translate into *output and inflation* responses.

11.1 Model architecture in one sentence

The framework is a New Keynesian economy (Rotemberg pricing) with a Poisson disaster risk and three household blocks (hand-to-mouth workers + two saver types with different subjective disaster intensities), where heterogeneous beliefs aggregate into a *market-implied disaster probability* that drives risk premia and feeds back into the macro dynamics.

11.2 The paper’s organizing logic

The analysis is structured around a sequence of implications:

1. Monetary shocks change the path of short rates, but also affect *discount rates* more broadly (including risk premia).
2. With disaster risk, asset prices respond through both *expected-rate components* and *risk-premium components*.
3. With heterogeneous beliefs, changes in discount rates redistribute wealth across saver types because their valuations differ.
4. With heterogeneous MPCs, redistribution becomes aggregate demand: *who gains wealth* matters more than *how much aggregate wealth* changes.
5. The **fiscal backing** of government liabilities (and the maturity structure of government debt) determines whether discount-rate movements translate into a *net* wealth effect for the private sector.

12 Conclusion

12.1 Monetary transmission is fundamentally shaped by risk premia

A key conclusion is that monetary policy shocks do not only operate through expected short-rate paths. They also operate through state-dependent *risk compensation*, and this risk-premium component is quantitatively important for matching the joint movements of:

- long-term yields and term premia,
- credit spreads (risk-premium component),
- equity valuations (discount-rate component),
- and macro outcomes (output and inflation).

12.2 Heterogeneity is the bridge from asset prices to real activity

Asset-price changes alone do not automatically create large consumption responses. The paper stresses that the real effects of asset revaluations depend on distribution:

- If wealth gains accrue to low-MPC agents, aggregate consumption reacts little.
- If policy-induced revaluation redistributes toward high-MPC agents (or away from them), aggregate demand responds strongly.

Thus, heterogeneity is not cosmetic: it is the mechanism that converts financial revaluation into macro dynamics.

12.3 Fiscal backing and government debt maturity are the hinge for “wealth effects”

A central message is that measured “wealth effects” from monetary policy are tightly linked to the government balance sheet:

- Long-maturity nominal debt makes the market value of government liabilities sensitive to discount-rate movements.
- The fiscal response (taxes/transfers) determines how much of that revaluation becomes private net wealth.

Hence, the wealth channel is jointly a *monetary–fiscal* phenomenon, not purely a monetary one.

12.4 Risk-premium neutrality clarifies interpretation of asset price movements

The paper shows that it is possible (under specific tax/transfer designs) to make macro outcomes largely invariant to belief-driven risk-premium movements, while allowing asset prices to move substantially. This delivers a sharp interpretation:

- Large asset-price responses to monetary policy *do not imply* large aggregate demand effects unless they create net wealth effects for high-MPC households.

12.5 Quantitatively, the model reduces the need for extreme EIS calibrations

In standard representative-agent NK models, large output responses often require extreme intertemporal substitution. By contrast, the paper concludes that introducing disaster risk and heterogeneity can generate realistic joint asset-price and macro responses with more plausible preference parameters, because the key amplification comes from:

risk premia + redistribution + high-MPC channels + fiscal backing.

13 Intuition

13.1 Intuition 1: “Risk” makes discount rates move for reasons beyond expected rates

With disaster risk, any monetary shock that changes economic conditions can shift the effective *price of tail risk* embedded in:

- term premia on long bonds,
- risk premia in corporate spreads,
- equity discount rates.

So, policy can tighten financial conditions not only by raising expected short rates, but also by raising the compensation investors demand for bearing disaster exposure.

13.2 Intuition 2: “Heterogeneous beliefs” turn discount-rate movements into redistribution

Optimists and pessimists price the same risky cash flows differently (because they disagree about disaster likelihood). When discount rates change, the market revalues claims in a way that shifts wealth across these groups. This generates a moving distribution of wealth even if aggregate resources do not jump.

13.3 Intuition 3: “Heterogeneous MPCs” convert redistribution into aggregate demand

Once wealth is reallocated, consumption responds according to MPC heterogeneity. Therefore, the sign and magnitude of the wealth channel depend on:

$$(\text{who holds duration / risky claims}) \times (\text{who has high MPC}).$$

This is why a model that matches asset-price dynamics can still fail to match macro dynamics unless it models heterogeneity correctly.

13.4 Intuition 4: “Fiscal backing” decides whether discount-rate changes create net wealth

Discount-rate changes revalue government liabilities. Whether households become wealthier depends on whether taxes/transfers offset this revaluation. Thus:

- strong backing can mute net wealth effects,
- weak or delayed backing can amplify them,
- debt maturity determines sensitivity to long-rate movements.

14 Takeaway

The paper’s bottom line is that monetary policy affects the real economy through a joint *risk-and-heterogeneity* channel: policy moves discount rates and risk premia; heterogeneous beliefs translate these movements into redistribution; heterogeneous MPCs map redistribution into aggregate demand; and the government balance sheet (debt maturity and fiscal backing) determines how much discount-rate variation becomes net private wealth. Consequently, interpreting monetary transmission requires an equilibrium framework that simultaneously matches asset pricing facts and cross-household heterogeneity rather than relying solely on representative-agent intertemporal substitution.

A Proofs (Reconstructed in a Self-Contained Way)

A.1 Proof of Proposition 1 (Market-implied disaster probability)

Proof. The risky-asset Euler conditions (2)–(3) imply that, for each saver type j , the relevant “price of disaster risk” is

$$\lambda_j \left(\frac{C_{j,t}}{C_{j,t}^*} \right)^\sigma.$$

To construct a single aggregate pricing kernel η_t that prices the same assets, we need a common object

$$\lambda_t \left(\frac{C_{s,t}}{C_{s,t}^*} \right)^\sigma$$

that matches each saver’s pricing of the jump component (because the jump is aggregate and identical across investors). Hence impose

$$\lambda_t \left(\frac{C_{s,t}}{C_{s,t}^*} \right)^\sigma = \lambda_j \left(\frac{C_{j,t}}{C_{j,t}^*} \right)^\sigma, \quad j \in \{o, p\}.$$

Rearrange to express disaster-state consumption of type j as

$$C_{j,t}^* = \left(\frac{\lambda_j}{\lambda_t} \right)^{1/\sigma} \frac{C_{s,t}^*}{C_{s,t}} C_{j,t}.$$

Plug this into the definition of average saver disaster consumption,

$$C_{s,t}^* \equiv \frac{\mu_o}{\mu_o + \mu_p} C_{o,t}^* + \frac{\mu_p}{\mu_o + \mu_p} C_{p,t}^*,$$

and solve for λ_t . This yields (4).

Finally, define the average saver discount rate $\rho_{s,t}$ so that the constructed aggregate SDF has drift consistent with pricing the risk-free bond:

$$\frac{\mathbb{E}_t[d\eta_t]}{\eta_t} = -(i_t - \pi_t)dt.$$

One convenient choice is

$$\rho_{s,t} \equiv \sum_{j \in \{o, p\}} \frac{\mu_j C_{j,t}}{\mu_o C_{o,t} + \mu_p C_{p,t}} (\rho_{j,t} + \lambda_j) - \lambda_t,$$

which pins down the correct drift. Therefore the aggregate SDF prices both risk-free and risky assets using the market-implied λ_t . $\square$

A.2 Proof of Proposition 2 (Aggregate dynamics)

Proof. (NKPC). Start from the nonlinear NKPC (5). Linearize around the stationary equilibrium using the steady-state condition $\frac{W}{P} = (1 - \varepsilon^{-1})A$ and the log-linear labor-supply relation $w_t - p_t = \phi y_t$. This yields

$$\dot{\pi}_t = (\rho_s + \lambda)\pi_t - \kappa y_t,$$

where $\kappa = \varphi^{-1}(\varepsilon - 1)\phi Y$. Set $\rho \equiv \rho_s + \lambda$ to obtain (11).

(Aggregate Euler). Goods market clearing implies

$$Y_t = \mu_w C_{w,t} + (\mu_o + \mu_p) C_{s,t}.$$

Log-linearizing and using worker consumption (7) gives

$$c_{s,t} = \frac{1 - \mu_w \chi_y}{1 - \mu_w} y_t.$$

Combine this with the saver aggregate Euler equation (8) and the stationary-equilibrium identity

$$r_n = \rho - \lambda \left(\frac{C_s}{C_s^*} \right)^\sigma$$

to obtain (10). Finally substitute $p_{d,t} = \tilde{\sigma} y_t + \hat{\lambda}_t$ to get the alternative representation with δ . $\square$

A.3 Proof of Proposition 3 (Approximate block-recursivity)

Proof. The (linearized) Euler equation for saver j can be written as

$$\dot{c}_{j,t} = \sigma^{-1}(i_t - \pi_t - r_n) + \frac{\lambda}{\sigma} \left(\frac{C_s}{C_s^*} \right)^\sigma (\hat{\lambda}_t + \sigma c_{s,t}) - \xi(c_{j,t} - c_{s,t}).$$

Taking the difference between types p and o cancels the common terms and yields

$$\dot{c}_{p,t} - \dot{c}_{o,t} = -\xi(c_{p,t} - c_{o,t}),$$

which is (16).

Next, linearize the saver flow budget constraint to obtain a law of motion for relative net worth $(b_{p,t} - b_{o,t})$ of the form

$$\dot{b}_{p,t} - \dot{b}_{o,t} = \rho(b_{p,t} - b_{o,t}) - \chi_{b,c}(c_{p,t} - c_{o,t}) + \chi_{b,c_s} c_{s,t},$$

where $\chi_{b,c} > 0$ and χ_{b,c_s} depends on portfolio differences and steady-state returns.

Under the stated second-order condition, the term involving $c_{s,t}$ can be neglected to first order

when solving for $(c_{p,t} - c_{o,t}, b_{p,t} - b_{o,t})$. The resulting 2×2 linear system has one stable and one unstable eigenvalue; imposing boundedness selects the unique stable solution:

$$b_{p,t} - b_{o,t} = e^{-\psi_\lambda t} (b_{p,0} - b_{o,0}), \quad \psi_\lambda = \xi,$$

and $c_{p,t} - c_{o,t}$ is proportional to $e^{-\psi_\lambda t} (b_{p,0} - b_{o,0})$.

Finally, log-linearizing λ_t using (15) shows $\hat{\lambda}_t$ is proportional to $(c_{p,t} - c_{o,t})$, hence

$$\hat{\lambda}_t = e^{-\psi_\lambda t} \hat{\lambda}_0.$$

If $i_t - r_n$ decays exponentially, bond revaluation pins down $(b_{p,0} - b_{o,0})$ as a linear function of $(i_0 - r_n)$, implying $\hat{\lambda}_0 = \varepsilon_\lambda (i_0 - r_n)$ with $\varepsilon_\lambda \geq 0$ and strictly positive when $\lambda_p > \lambda_o$. $\square$

A.4 Proof of Lemma 1 (iMPCs)

Proof. Household j satisfies the log-linear IBC (22):

$$\int_0^\infty e^{-\rho t} (c_{j,t} + \chi_{c_j} c_{j,t}^*) dt = \Omega_{j,0}.$$

Relative saver consumption satisfies the mean-reversion identity

$$\dot{c}_{j,t} = \dot{c}_{s,t} - \xi (c_{j,t} - c_{s,t}),$$

and the disaster vs no-disaster consumption gap satisfies a linear restriction of the form

$$\sigma (c_{j,t} - c_{j,t}^*) = \hat{\lambda}_t + \sigma (c_{s,t} - c_{s,t}^*).$$

Solving this linear system yields closed-form expressions for $c_{j,t}$ and $c_{j,t}^*$ as affine functions of $(\Omega_{j,0} - \Omega_{s,0})$ and $\hat{\lambda}_t$. Differentiating w.r.t. $\Omega_{j,0}$ gives

$$\mathcal{M}_{j,t} = \frac{\rho + \xi}{1 + \bar{\chi} \lambda_j^{1/\sigma}} e^{-\xi t}, \quad \mathcal{M}_{j,t}^* = \frac{(\rho + \xi) \bar{\chi}^* \lambda_j^{1/\sigma}}{1 + \bar{\chi} \lambda_j^{1/\sigma}} e^{-\xi t},$$

with positive constants $\bar{\chi}, \bar{\chi}^*$ determined by the steady state and the mapping between disaster-state and no-disaster-state consumption. The integral identity follows by differentiating (22) and using the relationship between χ_{c_j} and steady-state ratios such as δ/r_n^* . $\square$

A.5 Proof of Proposition 5 (Redistribution $\Rightarrow \hat{\lambda}_t$)

Proof. Start from the log-linearization (15), which gives

$$\hat{\lambda}_t = \chi_{\lambda,c}(c_{p,t} - c_{o,t}), \quad \text{for a constant } \chi_{\lambda,c} > 0 \text{ when } \lambda_p > \lambda_o.$$

Using the linear solution for saver consumptions from the iMPC lemma, the consumption gap $(c_{p,t} - c_{o,t})$ can be written as: (i) a term proportional to the difference in wealth effects $(\Omega_{p,0} - \Omega_{o,0})$ scaled by iMPC differences, plus (ii) a term proportional to $\hat{\lambda}_t$ itself. Solving this fixed-point representation yields

$$\hat{\lambda}_t = \bar{\chi}_\lambda (\mathcal{M}_{o,t} - \mathcal{M}_{p,t}) (\Omega_{p,0} - \Omega_{o,0}),$$

for a positive constant $\bar{\chi}_\lambda$. □

A.6 Proof of Propositions 6 and 7 (Output and inflation decompositions)

Proof. Write the linearized NK system in state vector form $Z_t \equiv (y_t, \pi_t)'$:

$$\dot{Z}_t = AZ_t + B\nu_t, \quad B = (1, 0)',$$

where ν_t collects the exogenous forcing terms (real-rate component and the risk component). The matrix A has two real eigenvalues $\bar{\omega} > 0$ and $\omega < 0$ under Assumption 1. Diagonalize $A = V\Lambda V^{-1}$ and solve the decoupled system forward for the unstable component and backward for the stable component, imposing boundedness.

This yields:

$$y_t = \tilde{y}_t + (\rho - \omega)e^{\omega t}\Omega_0, \quad \pi_t = \tilde{\pi}_t + \kappa e^{\omega t}\Omega_0,$$

where $\tilde{y}_t, \tilde{\pi}_t$ are linear functionals of ν_t .

With exponential monetary-shock decay and exponential $\hat{\lambda}_t$ decay, ν_t is a linear combination of $e^{-\psi_m t}$ and $e^{-\psi_\lambda t}$, which implies the closed-form $\hat{y}_{k,t}$ and $\hat{\pi}_{k,t}$ expressions in (27) and Proposition 7. Grouping coefficients delivers (26) and (28). □

A.7 Proof sketch for Proposition 8 (Household debt extension)

Proof. Worker wealth evolves with an additional private-bond position. Using the pricing relation for long-term private debt implies the yield component enters worker consumption as in (30). Goods market clearing then implies saver consumption depends on output and the private-yield and inflation terms.

Combining:

- saver Euler equation in aggregated form (with $\hat{\lambda}_t$),
- goods market clearing with private debt,
- exponential decay assumptions for $i_t - r_n$ and $\hat{\lambda}_t$,

yields an aggregate Euler equation with additional forcing terms proportional to $\bar{d}_P$. Solving the same 2×2 NK system via eigen-decomposition adds the “household debt effect” term stated in Proposition 8. $\square$